

FOOD_ORDERING_BEHAVIOUR AND CONSUMERED TRENDS: A STRUCTURED ANALYSIS OF ORDERS AND HABITS

TEAM ID	SWTID-2026-6508
PROJECT TITLE	FOOD_ORDERING_BEHAVIOUR AND CONSUMERED TRENDS : A STRUCTURED ANALYSIS OF ORDERS AND HABITS
MAX MARKS	5 MARKS

4.3 – Solution Architecture

The **Solution Architecture** for the project **"Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits"** is designed to provide an end-to-end framework for collecting, processing, analyzing, visualizing, and presenting food ordering data in an efficient and user-friendly manner. The architecture begins with the **data source layer**, where a structured dataset containing customer demographics, food ordering details, preferred cuisines, order frequency, spending patterns, payment methods, delivery preferences, restaurant ratings, discounts, and customer feedback is stored in **Microsoft Excel (.xlsx)** or **CSV** format. This data is then passed to the **data preprocessing layer**, where missing values, duplicate records, and inconsistencies are identified and corrected to ensure data quality and accuracy. The cleaned data is connected to **Tableau Desktop**, which forms the **analytics and visualization layer** of the architecture. Tableau processes the data and generates interactive worksheets, dashboards, charts, graphs, maps, and stories that provide detailed insights into customer behaviour, ordering habits, popular cuisines, spending trends, and customer satisfaction. These dashboards allow users to perform filtering, drill-down analysis, and dynamic exploration of the dataset. The next component is the **web application layer**, which is developed using **Python** and the **Flask** framework. Flask integrates the Tableau dashboards into a responsive website, enabling users to access and interact with the visualizations through a secure and intuitive

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interface. The front-end of the application is built using **HTML5**, **CSS3**, **Bootstrap**, and **JavaScript**, ensuring a modern, responsive, and user-friendly experience across different devices and browsers. Finally, the **presentation and decision-making layer** delivers meaningful reports and visual insights to business managers, restaurant owners, and stakeholders. These insights support strategic decisions such as optimizing menu offerings, improving delivery performance, identifying customer preferences, designing targeted marketing campaigns, and enhancing overall customer satisfaction. The architecture is modular, scalable, and easy to maintain, allowing future enhancements such as real-time data integration, predictive analytics, and machine learning capabilities. Overall, the proposed solution architecture provides a systematic flow of data from collection to visualization and decision-making, ensuring that the project effectively transforms raw food ordering data into actionable business intelligence that supports operational efficiency, customer engagement, and long-term growth in the online food delivery industry.

This architecture illustrates the complete flow of the project, from **data collection** to **analysis**, **web integration**, and **business decision-making**, ensuring a structured and efficient analytical solution.

BRAVO
— BUILDERS —

